

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Eighth Semester B.E. (Mechanical) (Credit System) Degree Examinations**

May - June 2023

19MEE42 – POWER PLANT ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I**Marks BT* CO* PO***

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|-------|--|----|-----|---|---|
| 1. a) | Explain the working of steam power plant with line diagram. What are the different circuits, major systems and components in the steam power plant? | 10 | L*1 | 1 | 1 |
| b) | Differentiate between the following.
i) Natural circulation and forced circulation boilers
ii) Forced draught and Induced Draught system. | 10 | L2 | 2 | 1 |
| 2. a) | List the boiler fuels used in steam power plant. Explain the important properties of coal required as a boiler fuel in steam power plant. | 10 | L2 | 1 | 1 |
| b) | Explain with neat sketch the working of Benson boiler and list its advantages. | 10 | L2 | 2 | 1 |
| 3. a) | List the advantages of pulverized coal firing. Also explain the unit system of pulverized fuel handling system. | 10 | L2 | 1 | 1 |
| b) | Mention the functions of the following in steam power plant:
i) Cooling tower
ii) Superheater
iii) Economizer
iv) Stack or chimney
v) Air preheater | 10 | L2 | 2 | 1 |

Unit – II

- | | | | | | |
|-------|--|----|----|---|---|
| 4. a) | List the major components in the Diesel Engine power plant. List the advantages and disadvantages of diesel engine power plant over steam power plant. | 10 | L1 | 3 | 1 |
| b) | Explain with neat sketch the working of PWR (Pressurized Water Reactor), also list its advantages and disadvantages. | 10 | L2 | 4 | 1 |

5. a) At a particular site following is mean monthly discharge. Draw hydrograph, flow duration curve and calculate the power available assuming overall efficiency 80% and there are 30 days in a month.

Month	Discharge (millions of m ³ /month)	Month	Discharge (millions of m ³ /month)
Apr	500	Oct	2000
May	200	Nov	1500
June	1500	Dec	1500
July	2500	Jan	1000
Aug	3000	Feb	800
Sept	2400	Mar	600

Head available is 80m.

10 L3 3 1

P.T.O.

- b) Differentiate between the open cycle and closed cycle gas turbine power plant. Also draw the line diagram of open cycle and closed cycle gas turbine power plant.

10 L2 4 1

6. a) What are the important functions of lubrication system in diesel engine power plant? Also explain the different methods of starting the diesel engine power plant.

10 L2 3 1

- b) Discuss the different methods of nuclear waste disposal in nuclear power plant.

10 L2 4 1

Unit – III

7. a) What is load duration curve? Mention the significance of Load duration curve.

4 L1 5 1

- b) Define the following.

i) Load factor

ii) Diversity factor

iii) Capacity factor

6 L1 5 1

- c) A residential load of a locality is given below. Draw the load curve and find load factor.

Time (hrs)	0-5	5-6	6-9	9-18	18-21	21-24
Load (MW)	2	6	20	0	12	8

10 L3 5 1

8. a) Calculate the cost of generation per kWh for a power station having the following data:

Installed capacity of the plant = 200 MW

Capital cost = Rs. 400 crores

Rate of interest and depreciation = 12%

Annual cost of fuel, salaries and taxation = Rs. 5 crores

Load factor = 50%

Also estimate the saving in cost per kWh if the annual load factor is raised to 60%

10 L3 5 1

- b) Discuss the various cost of electrical energy and factors in selecting the type of generation in power plant.

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
